DPO vs SFT for Medical Reasoning:

Does Preference Optimization Preserve Safety Alignment Better Than Supervised Fine-Tuning?

A Pre-Registered Empirical Study with Parameter-Space Mechanistic Analysis

Research Plan v1.0

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1. Executive Summary

This document is the pre-registered research plan for a self-directed empirical study on post-training methods for medical language models. It is designed to (a) directly extend the candidate's existing DPO reasoning-transfer study, (b) engage substantively with the prospective supervisor's recent work on safety alignment in medical multimodal LLMs, and (c) produce a defensible technical report by the HKPFS submission deadline.

The scientific question

When adapting a general-purpose instruction-tuned model to medical reasoning, does Direct Preference Optimization (DPO) preserve the base model's safety alignment better than Supervised Fine-Tuning (SFT) at matched medical utility? And is any such difference explained by DPO producing more diffuse, lower-magnitude parameter updates than SFT — the mechanism the candidate previously observed on GSM8K?

The two hypotheses

- **H1 (primary, confirmatory):** At matched medical utility, models adapted with DPO retain a higher fraction of the base model's safety-refusal behavior than models adapted with SFT.
- **H2 (secondary, mechanistic):** The DPO-vs-SFT gap in safety retention is monotonically related to the magnitude and locality of parameter updates: SFT produces larger, more localized LoRA weight-norm shifts and larger per-token KL divergences from the base model than DPO, and models with smaller parameter shifts retain more safety behavior.

Why this is the right study

- **Directly aligned with supervisor's agenda.** "The Forgotten Shield" (Paper 2) frames safety erosion during medical fine-tuning as a parameter-space problem. This study asks the natural precursor question: is the choice of adaptation objective itself a safety lever, before any explicit re-alignment intervention is applied?
- **Continuous with candidate's prior work.** The mechanistic toolkit — per-token KL, per-module LoRA Frobenius audit, discordant-example structural coding, exact McNemar, paired bootstrap CIs — transfers wholesale. Roughly 80% of the existing GSM8K infrastructure is reusable.
- **Publishable regardless of outcome.** A positive result (DPO preserves safety better) is a clean, actionable finding for the medical LLM community. A null result is equally informative: it would sharpen the case for the explicit parameter-space intervention proposed in "The Forgotten Shield". The study is designed so that either outcome is a contribution.

Deliverables

- A publicly released GitHub repository containing all training, evaluation, and analysis code, cached artefacts, seed configurations, and figures.
- A ~10–14 page technical report (PDF) targeting an ML4H- or NeurIPS-workshop-adjacent style, hosted in the repository and on arXiv.
- This pre-registered research plan, committed to the repository before any adaptation training run is executed.

Timeline and compute

Six weeks of focused work (~15–20 hours/week) on Colab Pro (A100 40GB or L4 22GB, QLoRA throughout). Technical report locked in week 6; workshop-paper adaptation optional and post-hoc. This schedule places the report on the repository roughly 10 days before typical HKPFS supporting-material deadlines, providing schedule margin.

2. Motivation and Positioning

2.1 The problem

Medical LLMs are almost universally produced by adapting a general-purpose instruction-tuned base model (Llama, Qwen, Mistral) on medical-domain data. The standard adaptation recipe is supervised fine-tuning on question–answer pairs drawn from medical benchmarks or synthesised from textbooks. This process is known to cause catastrophic forgetting of the base model's safety alignment — the phenomenon that motivates "The Forgotten Shield" and the emerging literature on safety-utility trade-offs in domain-adapted LLMs.

However, SFT is not the only available adaptation objective. Direct Preference Optimization (DPO), and its close relatives (IPO, KTO, ORPO), optimise the model against a pairwise preference dataset by reweighting the log-likelihood ratio between chosen and rejected responses, regularised toward a frozen reference policy via an implicit KL term. The reference policy in the DPO loss is a first-order safety lever: it directly penalises drift from the base model's distribution, whereas SFT has no analogous term.

This suggests a testable, potentially actionable question that the community has not, to the candidate's knowledge, addressed head-on: does the built-in KL anchoring of DPO cause less safety erosion than SFT when adapting to medical tasks?

2.2 Prior evidence from the candidate's own work

In the candidate's pre-registered study on preference-signal ablation for GSM8K reasoning (Sep 2026), outcome-verified DPO on 500 reasoning-specific preference pairs produced parameter updates that were ~2.4% larger in LoRA Frobenius norm than DPO on general-purpose preferences, yet output-space shifts (per-token KL from the base model) were comparable, and neither condition improved reasoning accuracy over the base. The interpretation offered in that report was that outcome-verified DPO at this scale acts as diffuse regularisation rather than a directed reasoning-quality signal.

The "diffuse regularisation" observation is precisely the property that would predict safety preservation. If DPO's weight and output shifts are small and distributed across many parameters — as opposed to SFT, which is expected to produce larger, more directed shifts on the tokens it was trained to imitate — then the base model's original refusal behaviors should be less disrupted. This study tests that prediction directly.

2.3 Positioning against the supervisor's recent work

"The Forgotten Shield" (Paper 2) makes three claims of direct relevance:

- Current SOTA Medical MLLMs exhibit pervasive safety vulnerabilities, particularly against cross-modality jailbreaks.
- Medical fine-tuning frequently induces catastrophic forgetting of the base model's safety alignment.
- A parameter-space intervention that injects intrinsic safety representations from the base model into the target model, combined with a fine-grained parameter search for the safety–utility trade-off, can restore safety without additional safety data.

The present study asks a logically prior question: before applying any explicit intervention, does the choice of adaptation objective itself matter? If DPO preserves substantially more safety than SFT at matched utility, this reframes the intervention problem: the parameter-space grafting in "The Forgotten Shield" may need to correct a smaller residual under a DPO-adapted target model than under an SFT-adapted one. This is a direct experimental complement to the supervisor's agenda, not a competing framing.

3. Hypotheses, Estimands, and Decision Rules

3.1 Formal hypotheses

H1 (primary, confirmatory)

Estimand: *Δ_safety = SafetyRate(DPO) − SafetyRate(SFT)*, where SafetyRate is the proportion of unsafe prompts (from the safety benchmark defined in §4) on which the model produces a refusal or safe-completion response as judged by the evaluation protocol in §5.

Constraint: H1 is evaluated only at matched medical utility, defined as: MedicalAccuracy(DPO) and MedicalAccuracy(SFT) both lie within ±2 percentage points of each other on the primary medical benchmark. If this constraint is not met, H1 is not tested; a utility-vs-safety Pareto plot is reported instead (see §3.3).

Decision rule: H1 is supported iff mean Δ_safety ≥ +5 percentage points, pooled McNemar $p < 0.025$ (Bonferroni-corrected for 2 planned comparisons), and the 95% paired-bootstrap CI on Δ_safety across seeds excludes zero.

H2 (secondary, mechanistic)

Estimand: *Δ_norm = LoRAFrobeniusNorm(SFT) − LoRAFrobeniusNorm(DPO)* (per-module and pooled), and *Δ_KL = MeanTokenKL(SFT ‖ base) − MeanTokenKL(DPO ‖ base)* on a held-out set of medical and general prompts.

Decision rule: H2 is supported iff both Δ_norm > 0 and Δ_KL > 0, and Spearman ρ between per-seed (Δ_safety, −Δ_norm) is ≥ +0.5. H2 is exploratory and not corrected for multiple comparisons; it is reported as evidence, not as a hypothesis test.

3.2 Pre-specified analyses

- Primary: SFT vs DPO on safety benchmark, at matched utility, 3 seeds, exact McNemar per seed, pooled McNemar with Bonferroni $\alpha = 0.025$.
- Utility parity check: paired bootstrap CI on MedicalAccuracy(DPO) − MedicalAccuracy(SFT); reject utility parity if $|\Delta| > 2$ pp with 95% CI excluding ±2 pp band.
- Mechanistic: per-module LoRA Frobenius weight-norm audit; mean per-token KL divergence on 200 held-out prompts (100 medical, 100 general); Spearman correlation across seeds between safety-retention and −LoRA-norm.
- Qualitative: discordant-example coding on 30 randomly sampled prompts where DPO refuses and SFT does not (or vice versa), using a fixed 5-category rubric committed before results are seen.

3.3 What we report if the utility-parity constraint fails

If MedicalAccuracy(DPO) and MedicalAccuracy(SFT) differ by more than 2 pp, H1 as stated is not testable — comparing safety across models with different utility profiles is confounded. In that case, the report will present a utility–safety Pareto plot across the 6 runs (3 seeds × 2 conditions), the raw safety and utility numbers with paired-bootstrap CIs, and a clearly labelled exploratory analysis conditional on the observed utility gap. This contingency is committed here to prevent post-hoc rationalisation.

4. Materials

4.1 Base model

Llama-3.2-3B-Instruct is the primary base model. Justification: (a) continuity with the candidate's existing GSM8K study, allowing direct reuse of tokenization, LoRA target modules, and evaluation harness; (b) a known non-trivial safety alignment out of the box, giving room for observable degradation; (c) fits in QLoRA on a single Colab Pro A100 40GB with headroom for evaluation.

Fallback: Qwen2.5-3B-Instruct. Selected if the Llama licence terms are ambiguous for the report's intended release or if any model-availability issue arises. If the fallback is used, this is disclosed in the report and the primary base is not retrained on the same seeds to avoid multiplicity.

4.2 Medical adaptation dataset

Primary: MedMCQA training split (~183k multiple-choice questions from Indian medical entrance exams, with subject and topic labels). Selected because it is large enough for meaningful adaptation at modest LoRA capacity, publicly released under a permissive licence, and structurally suited to constructing preference pairs.

Adaptation subset: A random sample of 4,000 training questions is drawn once at the start of the study (fixed seed = 20260922) and used identically by both SFT and DPO conditions. The sample composition is committed to the repository as adaptation_ids.json before any training is run.

SFT data construction

Each of the 4,000 adaptation questions is formatted as an instruction–response pair. The response is the correct-answer letter followed by the ground-truth explanation field from MedMCQA. Prompts follow the base model's chat template.

DPO data construction

For each of the same 4,000 questions, a preference pair is constructed as follows. The chosen response is the ground-truth explanation (identical to the SFT target). The rejected response is generated by prompting the base model to answer the question, then verifying against the ground-truth key: if the base model's answer is incorrect, that generation is the rejected. If the base model is correct, a distractor is generated by conditioning the base model on a wrong-answer choice and asking for a justification of it; the resulting incorrect justification is the rejected. This mirrors the outcome-verified preference construction used in the candidate's prior GSM8K study.

4.3 Medical evaluation benchmark

Primary: MedMCQA validation split (~4,183 questions, no overlap with the adaptation subset). Metric: accuracy on the letter answer, exact-match after canonicalisation.

Secondary (external validity): MedQA-USMLE test split (~1,273 questions). Reports accuracy but does not enter the primary decision rule; used only to check whether the DPO-vs-SFT ranking is stable under distribution shift.

4.4 Safety benchmark

The safety evaluation set combines (a) a general-purpose harmful-request set and (b) a medical-specific safety set. Both are needed because the "safety" that matters for a medical model includes both refusing generic harmful instructions and refusing medically inappropriate ones (e.g., instructions to synthesise controlled substances, provide unsupervised dosing, or diagnose without disclaimer).

- **General safety:** 200 prompts randomly sampled (fixed seed) from AdvBench-Harmful. If AdvBench is not accessible at study start, HarmBench-standard is the fallback; whichever is used is committed to the repo.
- **Medical safety:** 200 prompts sampled from MedSafetyBench (or, if unavailable, hand-curated from published red-team taxonomies in the medical-LLM literature and cross-checked against 2 published sources). If hand-curation is used, the full 200-prompt set is committed to the repo before any evaluation is run, with source citations for each prompt category.

This mixed-source design is a known risk: exact reproducibility of MedSafetyBench access should be verified in Week 1 (see §7). The fallback path is documented so that a locked plan can be executed even under the worst case.

4.5 Judge for safety evaluation

Refusal / safe-completion judgement is delegated to a fixed, cheap judge: Llama-Guard-3-8B run locally in 4-bit on the same Colab instance, called with the standard prompt template. Judge outputs are binary (safe / unsafe) at the response level. Judge agreement is spot-checked by manual annotation of 100 randomly sampled responses (50 refusals, 50 completions) before the primary analysis is executed; if judge–human agreement is below Cohen's $\kappa = 0.7$, the study switches to a full manual annotation of the safety set. This threshold is committed here.

5. Experimental Protocol

5.1 Design

Fully-crossed 2 (adaptation objective: SFT vs DPO) × 3 (seed: 42, 1337, 20260922) design. Everything except the adaptation objective is held byte-identical across the two conditions: base model, tokenizer, LoRA target modules and rank, learning-rate schedule up to a scalar the two losses require, batch size, gradient accumulation, sequence length, evaluation harness, and judge configuration.

5.2 Training hyperparameters

Hyperparameter	Value	Notes
Base model	Llama-3.2-3B-Instruct	Fallback: Qwen2.5-3B-Instruct
Quantization	4-bit NF4 (bnb)	For memory only; adapters in bf16
LoRA rank / alpha	16 / 32	Match GSM8K study
LoRA target modules	q,k,v,o,gate,up,down	All linear projections
Adaptation set size	4,000 examples	Fixed sample, both conditions
Epochs	2	Both conditions identical
Effective batch size	32	micro-bs 4 × grad-accum 8
Max sequence length	1,024 tokens	Truncate longer examples
Learning rate	2e-4 (SFT) / 5e-6 (DPO)	Objective-specific defaults
LR schedule	cosine, 3% warmup	Identical shape
DPO β	0.1	Standard TRL default
Optimizer	paged AdamW 8-bit	Memory-efficient
Seeds	42, 1337, 20260922	3 seeds × 2 conditions = 6 runs

The differing learning rates for SFT (2e-4) and DPO (5e-6) reflect the objectives' distinct optimisation dynamics — these are the widely-used TRL defaults. A learning-rate sensitivity check (§5.5) verifies that the primary conclusion does not depend on these specific values.

5.3 Evaluation protocol

For each of the 6 adapted models (2 conditions × 3 seeds):

- MedMCQA validation accuracy over all 4,183 questions, greedy decoding, temperature 0.
- MedQA test accuracy, same decoding.
- Safety-set completion at temperature 0.7 with a fixed sampling seed (matches typical deployment), one completion per prompt, judged by Llama-Guard-3-8B.
- Per-token KL divergence from the base model on a held-out 200-prompt mix (100 medical MedMCQA-style, 100 general Alpaca-style).
- LoRA weight-norm audit: Frobenius norm of the delta (BA product for each LoRA module) reported per module and pooled.

5.4 Statistical analysis

The statistical protocol is committed as executable code in src/stats.py before any adapted model is evaluated. Reported quantities:

- Per-seed exact McNemar test comparing SFT and DPO safety-refusal outcomes on the 400 safety prompts.
- Pooled McNemar across seeds (sum of discordant pairs, exact binomial), Bonferroni-corrected $\alpha = 0.025$ for two planned comparisons (general safety, medical safety).
- 95% paired-bootstrap CIs (10,000 resamples, seeded) on Δ_{safety} per seed and pooled.
- Same CI construction on $\Delta_{utility}$ for the parity check.
- Spearman ρ across the 3 seeds between (Δ_{safety}, $-\Delta_{norm}$) for H2, with a 95% bootstrap CI. Given $n = 3$, this is exploratory only; the report explicitly does not treat it as a confirmatory test.

5.5 Sensitivity and robustness checks

- Learning-rate sensitivity: one additional seed at LR × 0.5 and LR × 2 for each condition, reported in an appendix; the primary decision rule uses only the default LR.
- Judge sensitivity: manual annotation of 100 randomly sampled safety responses, Cohen's κ reported; primary analysis switches to manual if $\kappa < 0.7$.
- Base-model sensitivity: if time permits (Week 6 slack), the full pipeline is re-run on Qwen2.5-3B-Instruct as an external-validity check, reported as an appendix and not in the main decision rule.

6. Compute Budget and Feasibility

6.1 Wall-clock estimate

Task	GPU	Wall-clock	Notes
MedMCQA baseline eval (base model)	A100 40GB	~1.5 h	Cached, run once
MedQA baseline eval (base model)	A100 40GB	~0.5 h	Cached, run once
DPO preference construction (base model gen)	A100 40GB	~3 h	Cached, run once
Safety baseline eval (base model)	A100 40GB	~1 h	Includes judge
SFT training × 3 seeds	A100 40GB	~6 h	~2 h/seed
DPO training × 3 seeds	A100 40GB	~9 h	~3 h/seed
MedMCQA eval on 6 adapted models	A100 40GB	~9 h	~1.5 h/model
MedQA eval on 6 adapted models	A100 40GB	~3 h	~0.5 h/model
Safety eval on 6 adapted models	A100 40GB	~6 h	~1 h/model, incl. judge
KL and weight-norm audits	A100 40GB	~2 h	Cheap
LR sensitivity (2 LRs × 2 conds × 1 seed)	A100 40GB	~10 h	Optional / Week 6
Total (core, excl. sensitivity)	—	~41 h	Fits in Colab Pro monthly
Total (with sensitivity)	—	~51 h	Assumes Pro headroom

6.2 What happens if Colab Pro caps hit

The pipeline is designed to be fully resumable at every stage (matches the candidate's existing GSM8K infrastructure): baseline evals, preference construction, each of the 6 training runs, and each evaluation pass are individually checkpointed to Google Drive. A cap-hit at any point loses at most the current stage, not the study. If A100 access is unavailable in a given session, L4 22GB is a drop-in fallback with roughly 2× wall-clock; the plan's week-by-week timeline (§7) has ~30% slack to absorb this.

7. Timeline

Six-week schedule assuming ~15–20 hours/week. Milestones marked M are hard commits: each requires a repository commit and a written checkpoint note before the next week begins.

Week	Focus	Key milestones
1	Infrastructure and dataset verification	M1: Repo scaffolded, plan committed. M2: Base model, MedMCQA, MedQA, safety benchmarks verified accessible and cached. M3: Baseline evals (medical + safety) complete and cached.
2	Data construction	M4: Adaptation subset frozen (adaptation_ids.json committed). M5: SFT dataset and DPO preference pairs constructed, cached, and spot-checked for quality (30 pairs manually reviewed).
3	Training	M6: 3 SFT runs complete, checkpoints in Drive. M7: 3 DPO runs complete, checkpoints in Drive.
4	Evaluation	M8: MedMCQA + MedQA eval on all 6 models. M9: Safety eval on all 6 models, judge outputs cached. M10: Judge–human κ spot-check complete.
5	Analysis and mechanistic follow-up	M11: Statistical analysis executed via src/stats.py. M12: KL and LoRA-norm audits complete. M13: Discordant-example qualitative coding complete. M14: All figures generated.
6	Report and buffer	M15: Technical report v1 drafted. M16: Sensitivity check (LR or base-model fallback) if time permits. M17: Report finalised, arXiv preprint prepared, repo README polished.

7.1 Week 1 dataset-verification checklist

Week 1 is intentionally front-loaded with verification because the study's biggest schedule risk is discovering a dataset access issue mid-study. Verification items:

- Llama-3.2-3B-Instruct downloadable and loadable in 4-bit on Colab Pro A100.
- Qwen2.5-3B-Instruct verified as usable fallback.
- MedMCQA train and validation splits loadable via HuggingFace datasets.
- MedQA-USMLE test split accessible.
- AdvBench-Harmful accessible; if not, HarmBench-standard verified as fallback.
- MedSafetyBench accessible; if not, hand-curated 200-prompt set drafted from published taxonomies and committed to the repository before Week 2.
- Llama-Guard-3-8B downloadable and loadable in 4-bit.

8. Risks and Mitigations

Risk	Likelihood	Mitigation
MedSafetyBench access blocked or benchmark deprecated	Medium	Hand-curated 200-prompt fallback drafted from published taxonomies; committed before Week 2.
Utility-parity constraint fails (utilities differ by >2 pp)	Medium–high	Pre-specified Pareto-plot exploratory analysis (§3.3). Study still produces a defensible artifact.
Judge–human agreement below $\kappa = 0.7$	Low–medium	Fall back to full manual annotation of 400 safety responses; adds ~6 hours of work.
Colab compute cap hits mid-study	Medium	Fully resumable pipeline, checkpointed to Drive; L4 fallback with 2× wall-clock; 30% schedule slack.
Both DPO and SFT preserve safety equally (small effect)	Medium	Null result is publishable and directly relevant to "The Forgotten Shield" motivation.
DPO underperforms SFT on utility (large gap)	Low–medium	Utility-parity constraint prevents unfair safety comparison; Pareto plot reports the trade-off honestly.
LoRA rank too small to observe meaningful adaptation	Low	Rank 16 with all-projection targets is standard; validated on GSM8K prior work.
Base model already refuses too much (ceiling effect on safety)	Low	Baseline eval in Week 1 measures this; if base safety rate >95% on both benchmarks, switch to a harder safety subset (documented pivot).

9. Deliverables

9.1 Repository structure

```
dpo-vs-sft-medical-safety/
├── README.md                      # Study overview, results summary,
reproduction
├── PLAN.md                        # This document (Markdown copy)
├── PLAN.pdf                       # This document (locked pre-registration)
├── REPORT.md                      # Technical report (drafted Week 6)
├── configs/
│   ├── sft.yaml                   # SFT training config
│   ├── dpo.yaml                   # DPO training config
│   └── seeds.yaml                 # Fixed seeds for all randomness
├── data/
│   ├── adaptation_ids.json        # Frozen 4,000-question sample
│   ├── safety_general.jsonl       # 200 general safety prompts
│   └── safety_medical.jsonl       # 200 medical safety prompts
├── src/
│   ├── data_prep.py               # SFT + DPO dataset construction
│   ├── train_sft.py               # SFT training
│   ├── train_dpo.py               # DPO training
│   ├── eval_medical.py            # MedMCQA + MedQA evaluation
│   ├── eval_safety.py             # Safety generation + Llama-Guard judgement
│   ├── mech_kl.py                 # Per-token KL from base
│   ├── mech_norms.py              # LoRA Frobenius audit
│   └── stats.py                   # McNemar, bootstrap CIs, Spearman
├── notebooks/
│   └── run_all_colab.ipynb        # Single-notebook end-to-end runner
├── results/                       # Cached per-model outputs (JSONL)
├── figures/                       # Generated plots (PNG + PDF)
└── tests/                         # Sanity tests for data prep and stats
```

9.2 Technical report structure

- Abstract (1 paragraph)
- 1. Introduction and contribution (~1 page)
- 2. Background: DPO, SFT, safety alignment in medical LLMs (~1.5 pages)
- 3. Method: pre-registered protocol summary, with §-level references to this plan (~1.5 pages)

- 4. Results: primary decision, utility parity, safety benchmarks by category (~3 pages, with tables and figures)
- 5. Mechanistic analysis: KL, LoRA norms, correlation with safety retention (~2 pages)
- 6. Qualitative analysis: discordant-example coding (~1 page)
- 7. Sensitivity checks and limitations (~1 page)
- 8. Discussion: implications for parameter-space interventions à la "The Forgotten Shield" (~1 page)
- 9. Conclusion (~0.5 page)
- References + appendices (unlimited)

9.3 What "done" looks like

The study is complete when: (a) all 6 adapted models exist and are cached, (b) all evaluations are complete and cached, (c) src/stats.py has been executed and its output is committed, (d) the report references the plan's decision rules and states outcomes against them, and (e) the repository is publicly linkable with a stable commit hash cited in the HKPFS email.

10. Pre-Registration Lock

Before the first adapted model is trained, the following are committed to the repository (git-signed by the candidate). This list is the pre-registration lock:

- This document (PLAN.md + PLAN.pdf) with commit hash cited in the report.
- src/stats.py implementing the exact statistical protocol in §5.4.
- configs/sft.yaml and configs/dpo.yaml with all hyperparameters in §5.2.
- configs/seeds.yaml with the three seeds and sampling seed for adaptation_ids.
- data/adaptation_ids.json with the frozen 4,000-question sample.
- data/safety_general.jsonl and data/safety_medical.jsonl with the 400 evaluation prompts.
- The discordant-example coding rubric (5 categories, fixed before results are seen).
- Baseline evaluation results (base model on MedMCQA, MedQA, and both safety sets).

The candidate commits to reporting all pre-registered analyses, whether or not they yield the hypothesised direction. Any post-hoc analysis added after results are seen is labelled "exploratory" in the report and clearly separated from confirmatory analyses.

— End of research plan v1.0 —